MATA37H3 — Week 05

Improper Integrals: Type 1, Type 2, p-Integrals and the Comparison Tests

Recap and orientation

Last week completed the integration-techniques toolkit (trigonometric integrals, partial fractions, the Weierstrass substitution). This week those techniques are used inside **limits**: an **improper integral** extends $\int_a^b f(x)\,dx$ to an infinite interval (Type 1) or to an integrand that blows up at an endpoint or interior point (Type 2). Each is defined as a limit of proper integrals, and it **converges** only if that limit is finite.

Why this matters

When an antiderivative is hard or impossible to find, the **Basic Comparison Test** and **Limit Comparison Test** decide convergence by comparing with a benchmark p-integral. Series convergence tests in Weeks 6-14 are the discrete versions of exactly these arguments.

DEFINITION **Type-1 Improper Integral (Infinite Interval)**

If f is continuous on $[a,+\infty)$, then

$$\int_a^{+\infty} f(x)\,dx = \lim_{t\to+\infty}\int_a^t f(x)\,dx,$$

provided the limit exists. Similarly $\displaystyle\int_{-\infty}^b f(x)\,dx = \lim_{t\to-\infty}\int_t^b f(x)\,dx$. For $(-\infty,+\infty)$, split at any real a:

$$\int_{-\infty}^{+\infty} f(x)\,dx = \int_{-\infty}^a f(x)\,dx + \int_a^{+\infty} f(x)\,dx,$$

which converges only if **both** pieces converge. If the limit exists and is finite the integral **converges**; otherwise it **diverges**.

DEFINITION **Type-2 Improper Integral (Discontinuous Integrand)**

If f is continuous on $[a,b)$ with a discontinuity at b, then $\displaystyle\int_a^b f(x)\,dx = \lim_{t\to b^-}\int_a^t f(x)\,dx$. If the discontinuity is at a, use $\displaystyle\lim_{t\to a^+}\int_t^b f(x)\,dx$. If f is discontinuous at an interior point c, split:

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx,$$

which converges only if **both** pieces converge.

KEY FACT **The p-Integrals**

$$\int_1^{+\infty}\tfrac{1}{x^p}\,dx \text{ converges } \iff p>1, \qquad \int_0^1\tfrac{1}{x^p}\,dx \text{ converges } \iff p<1.$$

Consequently $\displaystyle\int_0^{+\infty} \frac{1}{x^p}\,dx$ diverges for every p, since it splits into the two integrals above and no single p makes both converge.

KEY FACT **Basic Comparison Test (BCT)**

Let f, g be continuous on $[a, +\infty)$ with $0 \le f(x) \le g(x)$ for all $x \ge a$.

- If $\displaystyle\int_a^{+\infty} f(x)\,dx$ diverges, then $\displaystyle\int_a^{+\infty} g(x)\,dx$ diverges ("small diverges $\Rightarrow$ big diverges").
- If $\displaystyle\int_a^{+\infty} g(x)\,dx$ converges, then $\displaystyle\int_a^{+\infty} f(x)\,dx$ converges ("big converges $\Rightarrow$ small converges").

The other two implications are false in general. The inequality only needs to hold eventually.

KEY FACT **Limit Comparison Test (LCT)**

Let f, g be positive and continuous on $[a, +\infty)$ with $L = \displaystyle\lim_{x\to+\infty} \frac{f(x)}{g(x)}$.

- If $0 < L < +\infty$, then $\displaystyle\int_a^{+\infty} f(x)\,dx$ and $\displaystyle\int_a^{+\infty} g(x)\,dx$ both converge or both diverge.
- The same statement holds for Type-2 integrals, taking the limit as x approaches the singular endpoint (for example $x \to 0^+$).

Worked Examples

EXAMPLE 1

Level: Foundational. Type 1 integral with partial fractions

Evaluate $\displaystyle\int_1^{+\infty} \frac{1}{x^2+x}\,dx$.

Step 1. Classify the integral. The upper limit is $+\infty$, so this is a Type 1 improper integral. By definition we replace the infinite limit by a finite number t and take a limit:

$$\int_1^{+\infty} \frac{1}{x^2+x}\,dx = \lim_{t\to+\infty} \int_1^t \frac{1}{x^2+x}\,dx.$$

The integrand is continuous on $[1, t]$ because the denominator $x^2 + x$ is positive there.

Step 2. To find an antiderivative, factor the denominator: $x^2 + x = x(x+1)$. The integrand is a rational function whose denominator is a product of distinct linear factors, so we use partial fractions:

$$\frac{1}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1}.$$

Step 3. Multiply both sides by the common denominator $x(x+1)$ to clear the fractions:

$1 = A(x+1) + Bx$. This equation must hold for every x.

Step 4. Substitute $x = 0$, which makes the B term vanish: $1 = A(0+1) + B \cdot 0 = A$, so $A = 1$.

Step 5. Substitute $x = -1$, which makes the A term vanish: $1 = A \cdot 0 + B(-1) = -B$, so $B = -1$.

Step 6. Check the decomposition by recombining: $\dfrac{1}{x} - \dfrac{1}{x+1} = \dfrac{(x+1) - x}{x(x+1)} = \dfrac{1}{x(x+1)}$. This equals the original integrand, so $\dfrac{1}{x^2+x} = \dfrac{1}{x} - \dfrac{1}{x+1}$.

Step 7. Integrate term by term. $\int \dfrac{1}{x}\,dx = \ln|x|$. For $\int \dfrac{1}{x+1}\,dx$ let $u = x+1$, so $du = dx$ and the integral is $\int \dfrac{du}{u} = \ln|u| = \ln|x+1|$. Therefore an antiderivative is $\ln|x| - \ln|x+1|$.

Step 8. On the interval $[1, t]$ both x and $x+1$ are positive, so the absolute value bars can be dropped:

$$\int_1^t \left(\frac{1}{x} - \frac{1}{x+1}\right) dx = \Big[\ln x - \ln(x+1)\Big]_1^t.$$

Step 9. Evaluate at the upper limit $x = t$: $\ln t - \ln(t+1)$. Evaluate at the lower limit $x = 1$: $\ln 1 - \ln 2 = 0 - \ln 2 = -\ln 2$.

Step 10. Subtract (upper minus lower):

$$\int_1^t \frac{1}{x^2+x}\,dx = \big(\ln t - \ln(t+1)\big) - \big(-\ln 2\big) = \ln t - \ln(t+1) + \ln 2.$$

Step 11. Combine the first two logarithms with the quotient rule $\ln a - \ln b = \ln \dfrac{a}{b}$:

$$\ln t - \ln(t+1) = \ln \frac{t}{t+1}.$$

Step 12. Take the limit $t \to +\infty$. Divide numerator and denominator of $\dfrac{t}{t+1}$ by t: $\dfrac{t}{t+1} = \dfrac{1}{1+\frac{1}{t}} \to \dfrac{1}{1+0} = 1$. Since ln is continuous at 1, $\ln \dfrac{t}{t+1} \to \ln 1 = 0$.

Step 13. Therefore the limit is $0 + \ln 2 = \ln 2$, which is finite, so the integral converges.

Step 14. Numerical check with $t = 1000$: $\ln \dfrac{1000}{1001} \approx -0.0009995$, so the partial integral is about $-0.0009995 + 0.693147 \approx 0.692148$, already very close to $\ln 2 \approx 0.693147$.

Answer. Converges to $\ln 2$

EXAMPLE 2

Level: Intermediate. The p-integral derived from the definition

Show from the definition that $\displaystyle\int_1^{+\infty} \frac{1}{x^p}\,dx$ converges if and only if $p > 1$, and find its value when it converges.

Step 1. Write the integral as a limit:

$$\int_1^{+\infty} \frac{1}{x^p}\,dx = \lim_{t\to+\infty} \int_1^t x^{-p}\,dx.$$

The antiderivative of x^{-p} depends on whether $-p = -1$, so we treat $p \neq 1$ and $p = 1$

separately.

Step 2. Case $p \neq 1$. Use the power rule $\int x^k\,dx = \dfrac{x^{k+1}}{k+1}$ with $k = -p$ (allowed because $k \neq -1$):

$$\int x^{-p}\,dx = \frac{x^{1-p}}{1-p}.$$

Check: $\dfrac{d}{dx}\left[\dfrac{x^{1-p}}{1-p}\right] = \dfrac{(1-p)x^{-p}}{1-p} = x^{-p}$.

Step 3. Evaluate from 1 to t, using $1^{1-p} = 1$:

$$\int_1^t x^{-p}\,dx = \frac{t^{1-p}-1}{1-p}.$$

Step 4. Subcase $p > 1$. Then $1-p < 0$, so write $1-p = -(p-1)$ with $p-1 > 0$. Hence $t^{1-p} = \dfrac{1}{t^{p-1}}$, and since $t^{p-1} \to +\infty$ as $t \to +\infty$, we get $t^{1-p} \to 0$.

Step 5. Then the limit is $\dfrac{0-1}{1-p} = \dfrac{-1}{1-p}$. Since $1-p = -(p-1)$, this equals $\dfrac{1}{p-1}$, a finite number. So for $p > 1$ the integral converges to $\dfrac{1}{p-1}$.

Step 6. Spot-check with $p = 2$: $\int_1^t x^{-2}\,dx = \left[-\dfrac{1}{x}\right]_1^t = 1 - \dfrac{1}{t} \to 1$, and the formula gives $\dfrac{1}{2-1} = 1$.

Step 7. Subcase $p < 1$. Then $1-p > 0$, so t^{1-p} is a positive power of t and $t^{1-p} \to +\infty$. The numerator $t^{1-p} - 1 \to +\infty$, and the denominator $1-p$ is a fixed positive number, so the quotient tends to $+\infty$. The integral diverges.

Step 8. Spot-check with $p = \dfrac{1}{2}$: $\int_1^t x^{-1/2}\,dx = \left[2\sqrt{x}\right]_1^t = 2\sqrt{t} - 2 \to +\infty$.

Step 9. Case $p = 1$. Here $\int \dfrac{1}{x}\,dx = \ln|x|$, so

$$\int_1^t \frac{1}{x}\,dx = \ln t - \ln 1 = \ln t.$$

Step 10. The function $\ln t$ is increasing and unbounded: for any bound M, $\ln t > M$ once $t > e^M$. So $\ln t \to +\infty$ and the integral diverges.

Step 11. Combine the cases: for $p > 1$ the limit is the finite number $\dfrac{1}{p-1}$; for $p < 1$ and for $p = 1$ the limit is $+\infty$. Therefore the integral converges exactly when $p > 1$.

Step 12. Final check with $p = 3$: the formula gives $\dfrac{1}{3-1} = \dfrac{1}{2}$, and directly $\left[-\dfrac{1}{2x^2}\right]_1^{+\infty} = 0 - \left(-\dfrac{1}{2}\right) = \dfrac{1}{2}$.

Answer. Converges if and only if $p > 1$, with value $\dfrac{1}{p-1}$; diverges for $p \leq 1$

EXAMPLE 3

Level: Intermediate. Both limits infinite: splitting at a point

Determine whether $\displaystyle\int_{-\infty}^{+\infty} x\,e^{-x^2}\,dx$ converges, and find its value if it does.

Step 1. Both limits of integration are infinite, so the definition requires a split at any convenient real number a:

$$\textstyle\int_{-\infty}^{+\infty} f(x)\,dx = \int_{-\infty}^{a} f(x)\,dx + \int_{a}^{+\infty} f(x)\,dx,$$

and the whole integral converges only if each of the two pieces converges on its own. Choose $a = 0$ because it simplifies the exponential.

Step 2. First find an antiderivative of xe^{-x^2}. The exponent $-x^2$ has derivative $-2x$, which is a constant multiple of the extra factor x, so use the substitution $u = -x^2$.

Step 3. Differentiate: $du = -2x\,dx$, so $x\,dx = -\dfrac{du}{2}$.

Step 4. Substitute: $\displaystyle\int xe^{-x^2}\,dx = -\frac{1}{2}\int e^u\,du = -\frac{1}{2}e^u = -\frac{1}{2}e^{-x^2}$.

Step 5. Check by differentiating with the chain rule: $\dfrac{d}{dx}\left[-\dfrac{1}{2}e^{-x^2}\right] = -\dfrac{1}{2}e^{-x^2}\cdot(-2x) = xe^{-x^2}$.

Step 6. Right piece. By definition $\displaystyle\int_0^{+\infty} xe^{-x^2}\,dx = \lim_{t\to+\infty}\int_0^t xe^{-x^2}\,dx$, and

$$\textstyle\int_0^t xe^{-x^2}\,dx = \Big[-\dfrac{1}{2}e^{-x^2}\Big]_0^t = -\dfrac{1}{2}e^{-t^2} - \left(-\dfrac{1}{2}e^0\right) = \dfrac{1}{2} - \dfrac{1}{2}e^{-t^2}.$$

Step 7. As $t\to+\infty$, $t^2\to+\infty$, so $e^{-t^2}\to 0$. The right piece converges to $\dfrac{1}{2}$.

Step 8. Left piece. By definition $\displaystyle\int_{-\infty}^0 xe^{-x^2}\,dx = \lim_{s\to-\infty}\int_s^0 xe^{-x^2}\,dx$, and

$$\textstyle\int_s^0 xe^{-x^2}\,dx = \Big[-\dfrac{1}{2}e^{-x^2}\Big]_s^0 = -\dfrac{1}{2} - \left(-\dfrac{1}{2}e^{-s^2}\right) = -\dfrac{1}{2} + \dfrac{1}{2}e^{-s^2}.$$

Step 9. As $s\to-\infty$, the square $s^2\to+\infty$ (a large negative number squared is a large positive number), so $e^{-s^2}\to 0$. The left piece converges to $-\dfrac{1}{2}$.

Step 10. Both pieces converge to finite numbers, so the full integral converges, and its value is the sum:

$$-\frac{1}{2} + \frac{1}{2} = 0.$$

Step 11. Consistency check with a different split point $a = 1$: $\displaystyle\int_1^{+\infty} = \lim_{t\to\infty}\left[-\frac{1}{2}e^{-x^2}\right]_1^t =$ $\dfrac{1}{2}e^{-1}$ and $\displaystyle\int_{-\infty}^1 = \left[-\frac{1}{2}e^{-x^2}\right]_{-\infty}^1 = -\frac{1}{2}e^{-1} - 0$. The sum is again 0.

Step 12. Check of the signs: the integrand $f(x) = xe^{-x^2}$ satisfies $f(-x) = -xe^{-x^2} = -f(x)$ (it is odd), so the left piece is the negative of the right piece, matching $-\dfrac{1}{2}$ and $+\dfrac{1}{2}$.

Answer. Converges, and $\displaystyle\int_{-\infty}^{+\infty} xe^{-x^2}\,dx = 0$

EXAMPLE 4

Level: Foundational. Type 2 integral with an endpoint singularity

Evaluate $\displaystyle\int_0^4 \frac{1}{\sqrt{4-x}}\,dx$.

Step 1. Locate any discontinuity. The integrand $\dfrac{1}{\sqrt{4-x}}$ is defined and continuous when $4-x>0$, that is for $x<4$. At $x=4$ the denominator is 0 and the function grows without bound (a vertical asymptote). So the discontinuity is at the upper endpoint and the integral is Type 2.

Step 2. By definition, stop short of the bad endpoint and take a limit:

$$\int_0^4 \frac{1}{\sqrt{4-x}}\,dx = \lim_{t\to 4^-}\int_0^t \frac{1}{\sqrt{4-x}}\,dx.$$

The notation $t\to 4^-$ means t approaches 4 from the left, which keeps t inside $[0,4)$.

Step 3. Rewrite the integrand with a negative exponent: $\dfrac{1}{\sqrt{4-x}} = (4-x)^{-1/2}$.

Step 4. Use the substitution $u=4-x$. Differentiating, $du=-dx$, so $dx=-du$.

Step 5. Substitute: $\int (4-x)^{-1/2}\,dx = -\int u^{-1/2}\,du$.

Step 6. Apply the power rule $\int u^k\,du = \dfrac{u^{k+1}}{k+1}$ with $k=-\dfrac{1}{2}$, so $k+1=\dfrac{1}{2}$:

$$\int u^{-1/2}\,du = \frac{u^{1/2}}{\frac{1}{2}} = 2\sqrt{u}.$$

Step 7. Include the minus sign from step 5 and return to x: $\displaystyle\int \frac{1}{\sqrt{4-x}}\,dx = -2\sqrt{u} = -2\sqrt{4-x}$.

Step 8. Check by differentiating: $\dfrac{d}{dx}\left[-2(4-x)^{1/2}\right] = -2\cdot\dfrac{1}{2}(4-x)^{-1/2}\cdot(-1) = (4-x)^{-1/2}$, which is the integrand.

Step 9. Evaluate on $[0,t]$: at the upper limit, $-2\sqrt{4-t}$; at the lower limit, $-2\sqrt{4-0} = -2\cdot 2 = -4$.

Step 10. Subtract:

$$\int_0^t \frac{1}{\sqrt{4-x}}\,dx = -2\sqrt{4-t} - (-4) = 4 - 2\sqrt{4-t}.$$

Step 11. Take the limit $t\to 4^-$. Then $4-t\to 0^+$, so $\sqrt{4-t}\to 0$, and the expression tends to $4-0=4$.

Step 12. The limit is finite, so the integral converges to 4, even though the integrand is unbounded near $x=4$.

Step 13. Numerical check: for $t=3.9999$, $4-t=0.0001$ and $\sqrt{0.0001}=0.01$, so the partial integral is $4-0.02=3.98$, close to 4.

Answer. Converges to 4

EXAMPLE 5

Level: Intermediate. Interior discontinuity: split at the bad point

Evaluate

$$\int_{-1}^{1} \frac{1}{x^3}\,dx,$$

or show that it diverges.

Step 1. Look for points where the integrand $f(x) = \dfrac{1}{x^3}$ fails to be continuous on $[-1, 1]$. The denominator x^3 is zero only at $x = 0$, and $\left|\dfrac{1}{x^3}\right| \to \infty$ as $x \to 0$. So f has a vertical asymptote at $x = 0$.

Step 2. The point $x = 0$ lies strictly inside the interval $(-1, 1)$, not at an endpoint. So this is a Type-2 improper integral with an interior discontinuity.

Step 3. The Fundamental Theorem of Calculus requires the integrand to be continuous on the whole closed interval, so it cannot be applied across $x = 0$. Instead we use the definition: split the integral at the discontinuity $c = 0$,

$$\int_{-1}^{1} \frac{1}{x^3}\,dx = \int_{-1}^{0} \frac{1}{x^3}\,dx + \int_{0}^{1} \frac{1}{x^3}\,dx,$$

and treat each piece as its own limit. The original integral converges only if both pieces converge; if either one diverges, the whole integral diverges.

Step 4. Take the first piece. On $[-1, 0)$ the function is continuous and the only trouble is at the right endpoint 0, so by the Type-2 definition

$$\int_{-1}^{0} \frac{1}{x^3}\,dx = \lim_{t \to 0^-} \int_{-1}^{t} x^{-3}\,dx.$$

Step 5. Find an antiderivative with the power rule $\int x^n\,dx = \dfrac{x^{n+1}}{n+1}$ for $n = -3$. Here $n + 1 = -2$, so $\int x^{-3}\,dx = \dfrac{x^{-2}}{-2} = -\dfrac{1}{2x^2}$.

Step 6. Check the antiderivative by differentiating: $\dfrac{d}{dx}\left[-\dfrac{1}{2}x^{-2}\right] = -\dfrac{1}{2} \cdot (-2)x^{-3} = x^{-3}$, which is the integrand.

Step 7. Evaluate between the limits -1 and t (note $t < 0$, but $t^2 > 0$):

$$\int_{-1}^{t} x^{-3}\,dx = \left[-\frac{1}{2x^2}\right]_{-1}^{t} = -\frac{1}{2t^2} - \left(-\frac{1}{2(-1)^2}\right) = -\frac{1}{2t^2} + \frac{1}{2}.$$

Step 8. Take the limit $t \to 0^-$. Then $t^2 \to 0^+$, so $\dfrac{1}{2t^2} \to +\infty$ and therefore $-\dfrac{1}{2t^2} + \dfrac{1}{2} \to -\infty$.

Step 9. The first piece diverges (to $-\infty$). By the definition in step 3, a single divergent piece makes the whole improper integral diverge.

Step 10. For completeness, the second piece also diverges: $\lim\limits_{t \to 0^+} \displaystyle\int_{t}^{1} x^{-3}\,dx = \lim\limits_{t \to 0^+} \left[-\dfrac{1}{2x^2}\right]_{t}^{1} = \lim\limits_{t \to 0^+} \left(-\dfrac{1}{2} + \dfrac{1}{2t^2}\right) = +\infty$. The two infinite contributions $-\infty$ and $+\infty$ do not cancel; each piece must converge on its own.

Step 11. Cross-check with the p-integral rule $\int_0^1 \dfrac{1}{x^p}\,dx$ converges if and only if $p < 1$. Here $p = 3$, which is not less than 1, so the piece on $[0, 1]$ diverges, in agreement with the computation above (and the same holds for the mirror piece on $[-1, 0]$).

Step 12. Conclusion: $\displaystyle\int_{-1}^{1} \frac{1}{x^3}\,dx$ diverges.

Answer. The integral diverges: $\displaystyle\int_{-1}^{0} \frac{1}{x^3}\,dx = -\infty$ and $\displaystyle\int_{0}^{1} \frac{1}{x^3}\,dx = +\infty$.

EXAMPLE 6

Level: Intermediate. Basic Comparison Test: small diverges, so big diverges

Determine whether

$$\int_1^{+\infty} \frac{1+\cos^2 x}{x^{\frac{2}{3}}}\,dx$$

converges or diverges.

Step 1. The antiderivative of $\dfrac{1+\cos^2 x}{x^{2/3}}$ is not available in closed form, so we cannot use the definition directly. We compare the integrand with a simpler function whose integral we can compute. Write $f(x) = \dfrac{1+\cos^2 x}{x^{\frac{2}{3}}}$.

Step 2. Check that f is continuous and nonnegative on $[1, +\infty)$. The numerator $1+\cos^2 x$ is continuous, and the denominator $x^{\frac{2}{3}}$ is continuous and at least 1 for $x \geq 1$, so it never vanishes. Also $1+\cos^2 x > 0$, so $f(x) > 0$.

Step 3. Bound the numerator from below. A square is never negative, so $\cos^2 x \geq 0$ for every real x. Adding 1 to both sides gives $1+\cos^2 x \geq 1$.

Step 4. Divide both sides of $1+\cos^2 x \geq 1$ by $x^{\frac{2}{3}}$. Because $x^{\frac{2}{3}} > 0$ for $x \geq 1$, dividing by it keeps the inequality direction:

$$f(x) = \frac{1+\cos^2 x}{x^{\frac{2}{3}}} \geq \frac{1}{x^{\frac{2}{3}}} = g(x) > 0 \qquad \text{for all } x \geq 1.$$

Step 5. So $0 \leq g(x) \leq f(x)$ on $[1, +\infty)$, with $g(x) = x^{-\frac{2}{3}}$ continuous. This is the setting of the Basic Comparison Test, with g playing the role of the smaller function.

Step 6. Decide the behaviour of $\displaystyle\int_1^{+\infty} g(x)\,dx$. This is a p-integral with $p = \frac{2}{3}$, and $\frac{2}{3} \leq 1$, so it should diverge. Confirm from the definition with the power rule $\int x^n\,dx = \dfrac{x^{n+1}}{n+1}$, $n = -\dfrac{2}{3}$, $n+1 = \dfrac{1}{3}$:

$$\int_1^t x^{-\frac{2}{3}}\,dx = \left[\frac{x^{\frac{1}{3}}}{\frac{1}{3}}\right]_1^t = \left[3x^{\frac{1}{3}}\right]_1^t = 3t^{\frac{1}{3}} - 3.$$

Step 7. Check the antiderivative: $\dfrac{d}{dx}\left[3x^{\frac{1}{3}}\right] = 3 \cdot \dfrac{1}{3}x^{-\frac{2}{3}} = x^{-\frac{2}{3}}$.

Step 8. Take the limit as $t \to +\infty$: $t^{\frac{1}{3}} \to +\infty$, so $3t^{\frac{1}{3}} - 3 \to +\infty$. Hence $\displaystyle\int_1^{+\infty} \frac{1}{x^{\frac{2}{3}}}\,dx$ diverges.

Step 9. State the Basic Comparison Test, part (i): if $0 \le g(x) \le f(x)$ on $[a, +\infty)$ and $\displaystyle\int_a^{+\infty} g(x)\,dx = +\infty$, then $\displaystyle\int_a^{+\infty} f(x)\,dx = +\infty$ (the smaller function diverges, so the larger one diverges).

Step 10. Make the conclusion concrete without quoting the theorem: for every $t \ge 1$, $\displaystyle\int_1^t f(x)\,dx \ge \int_1^t g(x)\,dx = 3t^{\frac{1}{3}} - 3$, and the right side tends to $+\infty$. A quantity that is at least something tending to $+\infty$ also tends to $+\infty$.

Step 11. Spot check of the inequality at $x = 1$: $f(1) = \dfrac{1 + \cos^2 1}{1} \approx 1.292$ and $g(1) = 1$, so $f(1) \ge g(1)$ as claimed.

Step 12. Conclusion: $\displaystyle\int_1^{+\infty} \frac{1 + \cos^2 x}{x^{\frac{2}{3}}}\,dx$ diverges.

Answer. The integral diverges (it is bounded below by $\displaystyle\int_1^{+\infty} \frac{1}{x^{\frac{2}{3}}}\,dx$, a divergent p-integral with $p = \dfrac{2}{3}$).

EXAMPLE 7

Level: Challenge. Basic Comparison Test with a logarithm: a fully justified bound

Determine whether

$$\int_2^{+\infty} \frac{(\ln x)^{10}}{x^2}\,dx$$

converges or diverges.

Step 1. Write $f(x) = \dfrac{(\ln x)^{10}}{x^2}$. For $x \ge 2$ we have $\ln x \ge \ln 2 > 0$ and $x^2 > 0$, so f is continuous and positive on $[2, +\infty)$. There is no elementary antiderivative that is convenient, so we use a comparison.

Step 2. The logarithm grows more slowly than any positive power of x, so we expect $(\ln x)^{10}$ to be dominated by a small power such as $x^{\frac{1}{2}}$, leaving a p-integral with $p = 2 - \dfrac{1}{2} = \dfrac{3}{2} > 1$. We now prove a bound valid for every $x \ge 2$ (no unproved phrase such as 'for large enough x').

Step 3. Lemma: $\ln y \le y - 1$ for every $y > 0$. Proof: let $h(y) = y - 1 - \ln y$. Then

$h'(y) = 1 - \dfrac{1}{y}$, which is negative for $0 < y < 1$ and positive for $y > 1$, so h has its minimum at $y = 1$, where $h(1) = 1 - 1 - 0 = 0$. Hence $h(y) \geq 0$ for all $y > 0$, that is $\ln y \leq y - 1 < y$.

Step 4. Apply the lemma with $y = x^{\frac{1}{20}}$, which is positive. By the logarithm power rule $\ln\left(x^{\frac{1}{20}}\right) = \dfrac{1}{20}\ln x$, so $\dfrac{1}{20}\ln x \leq x^{\frac{1}{20}}$. Multiply by 20:

$$\ln x \leq 20\, x^{\frac{1}{20}} \qquad \text{for all } x > 0.$$

Step 5. Raise both sides to the power 10. For $x \geq 2$ both sides are positive ($0 < \ln x \leq 20\, x^{\frac{1}{20}}$), and $t \mapsto t^{10}$ is increasing on positive numbers, so the inequality is preserved. Using $(ab)^{10} = a^{10}b^{10}$ and $\left(x^{\frac{1}{20}}\right)^{10} = x^{\frac{10}{20}} = x^{\frac{1}{2}}$:

$$(\ln x)^{10} \leq 20^{10}\, x^{\frac{1}{2}} \qquad \text{for all } x \geq 2.$$

Step 6. Divide by $x^2 > 0$ and use the exponent rule $\dfrac{x^{\frac{1}{2}}}{x^2} = x^{\frac{1}{2}-2} = x^{-\frac{3}{2}}$:

$$0 < f(x) = \frac{(\ln x)^{10}}{x^2} \leq \frac{20^{10}}{x^{\frac{3}{2}}} = g(x) \qquad \text{for all } x \geq 2.$$

The bound holds on the entire interval, so no 'eventually' argument is needed. The constant 20^{10} is huge, but a constant factor does not affect convergence.

Step 7. Now compute the comparison integral from the definition. By the power rule with $n = -\dfrac{3}{2}$, $n + 1 = -\dfrac{1}{2}$:

$$\int_2^t x^{-\frac{3}{2}}\, dx = \left[\frac{x^{-\frac{1}{2}}}{-\frac{1}{2}}\right]_2^t = \left[-\frac{2}{\sqrt{x}}\right]_2^t = -\frac{2}{\sqrt{t}} + \frac{2}{\sqrt{2}} = \sqrt{2} - \frac{2}{\sqrt{t}}.$$

Step 8. Check the antiderivative: $\dfrac{d}{dx}\left[-2x^{-\frac{1}{2}}\right] = -2 \cdot \left(-\dfrac{1}{2}\right) x^{-\frac{3}{2}} = x^{-\frac{3}{2}}$.

Step 9. Let $t \to +\infty$: $\dfrac{2}{\sqrt{t}} \to 0$, so $\displaystyle\int_2^{+\infty} x^{-\frac{3}{2}}\, dx = \sqrt{2}$, which is finite. Therefore $\displaystyle\int_2^{+\infty} g(x)\, dx = 20^{10}\sqrt{2}$ converges (a finite constant times a convergent integral).

Step 10. State the Basic Comparison Test, part (ii): if $0 \leq f(x) \leq g(x)$ on $[a, +\infty)$ and

$\int_a^{+\infty} g(x)\,dx$ converges, then $\int_a^{+\infty} f(x)\,dx$ converges (the bigger function converges, so the smaller one converges).

Step 11. Concretely, $F(t) = \int_2^t f(x)\,dx$ is increasing in t (because $f > 0$) and bounded above by $20^{10}\sqrt{2}$ for every $t \geq 2$. An increasing function of t that is bounded above has a finite limit as $t \to +\infty$ (monotone convergence), so the improper integral converges.

Step 12. Conclusion: $\int_2^{+\infty} \frac{(\ln x)^{10}}{x^2}\,dx$ converges, and its value is at most $20^{10}\sqrt{2}$.

Answer. The integral converges, since $0 < \frac{(\ln x)^{10}}{x^2} \leq \frac{20^{10}}{x^{\frac{3}{2}}}$ for all $x \geq 2$ and $\int_2^{+\infty} \frac{dx}{x^{\frac{3}{2}}} = \sqrt{2}$.

EXAMPLE 8

Level: Advanced. Limit Comparison Test

Determine whether

$$\int_1^{+\infty} \frac{x+2}{\sqrt{x^4+x+1}}\,dx$$

converges or diverges.

Step 1. Write $f(x) = \frac{x+2}{\sqrt{x^4+x+1}}$. For $x \geq 1$ the numerator is positive and the radicand $x^4 + x + 1 \geq 3$ is positive, so f is continuous and positive on $[1, +\infty)$.

Step 2. A pointwise inequality such as $f(x) \leq \frac{1}{x}$ is awkward to prove here because of the square root, so use the Limit Comparison Test. First find the benchmark by keeping only the dominant term in each part for large x: the numerator behaves like x, and $\sqrt{x^4+x+1}$ behaves like $\sqrt{x^4} = x^2$.

Step 3. So $f(x)$ behaves like $\frac{x}{x^2} = \frac{1}{x}$. Choose $g(x) = \frac{1}{x}$, which is continuous and positive on $[1, +\infty)$.

Step 4. Form the ratio and simplify the complex fraction by multiplying by the reciprocal of $\frac{1}{x}$, which is x:

$$\frac{f(x)}{g(x)} = \frac{x+2}{\sqrt{x^4+x+1}} \cdot x = \frac{x^2+2x}{\sqrt{x^4+x+1}}.$$

Step 5. Factor the highest power out of the numerator: $x^2 + 2x = x^2\left(1 + \frac{2}{x}\right)$.

Step 6. Factor the highest power out of the radicand: $x^4 + x + 1 = x^4\left(1 + \frac{1}{x^3} + \frac{1}{x^4}\right)$. Since $x > 0$, $\sqrt{x^4} = x^2$, so $\sqrt{x^4+x+1} = x^2\sqrt{1 + \frac{1}{x^3} + \frac{1}{x^4}}$.

Step 7. Substitute and cancel the common factor x^2:

$$\frac{f(x)}{g(x)} = \frac{x^2\left(1+\dfrac{2}{x}\right)}{x^2\sqrt{1+\dfrac{1}{x^3}+\dfrac{1}{x^4}}} = \frac{1+\dfrac{2}{x}}{\sqrt{1+\dfrac{1}{x^3}+\dfrac{1}{x^4}}}.$$

Step 8. Take the limit as $x \to +\infty$. Each of $\dfrac{2}{x}$, $\dfrac{1}{x^3}$, $\dfrac{1}{x^4}$ tends to 0, and the square root is continuous at 1, so

$$L = \lim_{x\to+\infty} \frac{f(x)}{g(x)} = \frac{1+0}{\sqrt{1+0+0}} = 1.$$

Step 9. Numerical spot check at $x = 1000$: $f(1000) = \dfrac{1002}{\sqrt{10^{12}+1001}} \approx \dfrac{1002}{1.0000000005\times 10^6} \approx 1.002\times 10^{-3}$, and $g(1000) = 10^{-3}$, so the ratio is about 1.002, close to $L = 1$.

Step 10. State the Limit Comparison Test: if $f, g \geq 0$ are continuous on $[a, +\infty)$ and $\lim\limits_{x\to+\infty} \dfrac{f(x)}{g(x)} = L$ with $0 < L < \infty$, then $\displaystyle\int_a^{+\infty} f(x)\,dx$ and $\displaystyle\int_a^{+\infty} g(x)\,dx$ either both converge or both diverge. Here $L = 1$ is finite and nonzero, so the test applies.

Step 11. Decide the benchmark integral: $\displaystyle\int_1^t \frac{1}{x}\,dx = \ln t - \ln 1 = \ln t \to +\infty$ as $t \to +\infty$. So $\displaystyle\int_1^{+\infty} \frac{1}{x}\,dx$ diverges (it is the p-integral with $p = 1$).

Step 12. Conclusion: since the benchmark diverges and the two integrals behave the same way, $\displaystyle\int_1^{+\infty} \frac{x+2}{\sqrt{x^4+x+1}}\,dx$ diverges.

Answer. The integral diverges (LCT with $g(x) = \dfrac{1}{x}$, $L = 1$).

Practice Problems

Computing from the definition

1. Evaluate $\displaystyle\int_2^{+\infty} \frac{1}{(x-1)^2}\,dx$, or show it diverges.

Your working:

2. Evaluate $\displaystyle\int_0^{+\infty} x\, e^{-2x}\, dx$, or show it diverges.

Your working:

3. Evaluate $\displaystyle\int_{-\infty}^{0} \frac{1}{x^2+4}\, dx$, or show it diverges.

Your working:

Type-2 and split integrals

4. Evaluate $\displaystyle\int_0^{1} \frac{1}{\sqrt{x}}\, dx$, or show it diverges.

Your working:

5. Determine whether $\displaystyle\int_{-8}^{8} \frac{1}{x^{2/3}}\,dx$ converges or diverges, and find its value if it converges.

Your working:

6. A student evaluates $\displaystyle\int_{0}^{2} \frac{1}{(x-1)^2}\,dx$ as follows: $\displaystyle\int_{0}^{2} (x-1)^{-2}\,dx = \left[-\frac{1}{x-1}\right]_{0}^{2} = -1 - 1 = -2$. Explain precisely what is wrong with this computation, and determine the correct answer (a convergent value, or divergence).

Your working:

Rapid p-integral and comparison judgment

7. State whether

$$\int_1^{+\infty} \frac{1}{x^5}\,dx$$

converges or diverges, with one line of justification.

Your working:

8. State whether

$$\int_0^1 \frac{1}{x^5}\,dx$$

converges or diverges, with one line of justification.

Your working:

9. Use the Basic Comparison Test to decide whether

$$\int_1^{+\infty} \frac{2+\sin x}{x^{\frac{3}{4}}}\,dx$$

converges or diverges.

Your working:

10. Determine whether

$$\int_0^{+\infty} \frac{1}{x^{\frac{3}{2}}}\,dx$$

converges or diverges.

Your working:

BCT / LCT applications

11. Use the Basic Comparison Test to determine whether

$$\int_1^{+\infty} \frac{\arctan\left(x^2\right)}{1+x^3}\,dx$$

converges or diverges.

Your working:

12. Use the Limit Comparison Test to determine whether

$$\int_1^{+\infty} \frac{x^3+2x+7}{x^5+11x^4+1}\,dx$$

converges or diverges.

Your working:

13. Use the Limit Comparison Test with $g(x) = \dfrac{1}{x}$ to determine whether

$$\int_2^{+\infty} \frac{\ln x}{x+1}\, dx$$

converges or diverges.

Your working:

14. Use the Limit Comparison Test to determine whether

$$\int_0^1 \frac{\sin x}{x^{\frac{3}{2}}}\, dx$$

converges or diverges.

Your working:

15. Use the Limit Comparison Test to determine whether

$$\int_0^1 \frac{1}{\sqrt{1-x^3}}\,dx$$

converges or diverges.

Your working:

Applications

16. A random variable X has probability density $f(x) = \dfrac{k}{x^4}$ for $x \geq 2$ and $f(x) = 0$ for $x < 2$. Find the constant k that makes f a valid density, and then find $P(X > 4)$.

Your working:

17. The gravitational force on a satellite of mass m at distance r from the centre of the Earth (mass M) is $F(r) = \dfrac{GMm}{r^2}$, where G is the gravitational constant. Find the work needed to move the satellite from the distance $r = R$ (with $R > 0$) to arbitrarily far away.

Your working:

18. A damped oscillator moves along a line with velocity $v(t) = e^{-2t}\sin(3t)$ metres per second for $t \geq 0$. Find its total displacement $\displaystyle\int_0^{+\infty} e^{-2t}\sin(3t)\,dt$.

Your working:

19. The lifetime X (in thousands of hours) of a certain component has probability density function

$$f(x) = \begin{cases} \dfrac{k}{x^3}, & x \geq 2, \\ 0, & x < 2. \end{cases}$$

(a) Find the constant k. (b) Find $P(X > 4)$.

Your working:

20. The integral $\Gamma(3) = \displaystyle\int_0^{+\infty} x^2 e^{-x}\, dx$ appears in the gamma function, which extends the factorial to non-integers. Evaluate it, or show that it diverges.

Your working:

Error diagnosis

21. A student claims that $\displaystyle\int_1^{+\infty} \frac{1}{x}\,dx$ converges, because $\dfrac{1}{x} \to 0$ as $x \to +\infty$, so the area under the curve "runs out". Find the error and give the correct result.

Your working:

22. A student claims that $\displaystyle\int_{-\infty}^{+\infty} \frac{x}{x^2+1}\,dx = 0$, reasoning: "the integrand is odd, so $\displaystyle\int_{-t}^{t} \frac{x}{x^2+1}\,dx = 0$ for every t, hence the limit as $t \to +\infty$ is 0." Find the error and give the correct result.

Your working: